

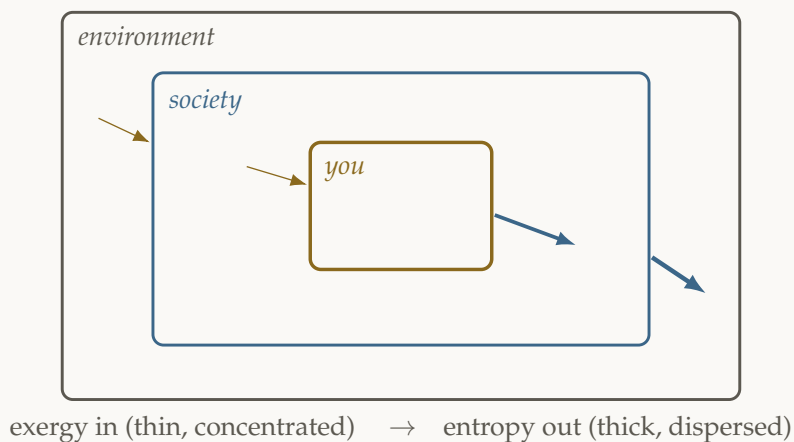
No View From Outside

Problems from inside the sandwich

A companion worksheet to *The Sandwich in the Middle*

Name: Class: Date:

A power station is easy to draw as a box with an arrow of fuel going in and an arrow of waste coming out, because nobody imagines the person drawing the diagram is standing inside it. That picture breaks down the moment the box is you, your home, or your school, since each of these is already inside the flow of energy it is trying to measure. The diagram below nests three such boxes inside one another; the arrows crossing each boundary are the same two arrows, concentrated energy in, dispersed waste out, drawn at three different scales.



A Your share of the waste

Before you calculate: Guess, before working it out, whether your personal share of the world's energy use is bigger or smaller than the heat your own body gives off just by staying alive.

World primary energy use is about 1.9×10^{13} W, shared across a world population of about 8.1×10^9 people. A resting human body gives off heat, mostly from its own metabolism, at a rate of roughly 100 W.

1. Calculate the average power used worldwide per person, in watts, to 2 significant figures.
2. Express your answer to A1 as a multiple of the 100 W a resting body radiates, to 1 decimal place.
3. In one or two sentences, say what your answer to A2 tells you about the difference between you as a biological system and you as a participant in the wider economic system.

B The heat pump on your wall

A heat pump does not burn anything. It uses a small amount of electrical work to drag heat out of the cold air outside and push it into your warm home, which is exactly a Carnot engine run backwards. Its performance is measured by a coefficient of performance, COP, the heat delivered indoors for every unit of electrical work put in, and the best possible COP obeys the same law as the power station's efficiency, only inverted:

$$\text{COP}_{\text{max}} = \frac{T_{\text{hot}}}{T_{\text{hot}} - T_{\text{cold}}} \quad (\text{temperatures in kelvin})$$

Before you calculate: Guess whether a heat pump can ever deliver more heat energy than the electrical energy it consumes, and if so, how that squares with energy being conserved.

1. Outside air is at 2°C and the heat pump delivers heat to radiators at 45°C. Convert both temperatures to kelvin.
2. Calculate COP_{max} for these temperatures.
3. A real domestic heat pump running between these temperatures achieves a COP of about 3. What percentage of the theoretical maximum is this?
4. Explain, in a sentence or two, why a COP greater than 1 does not break the first law of thermodynamics.

C Compounding, at your scale and at the world's

The rule of 70 gives a quick estimate for how long anything growing steadily takes to double:

$$\text{doubling time (years)} \approx \frac{70}{\text{percentage growth rate per year}}$$

Before you calculate: Guess which doubles faster: a household electricity bill rising 6% a year, or world energy use rising 2.3% a year.

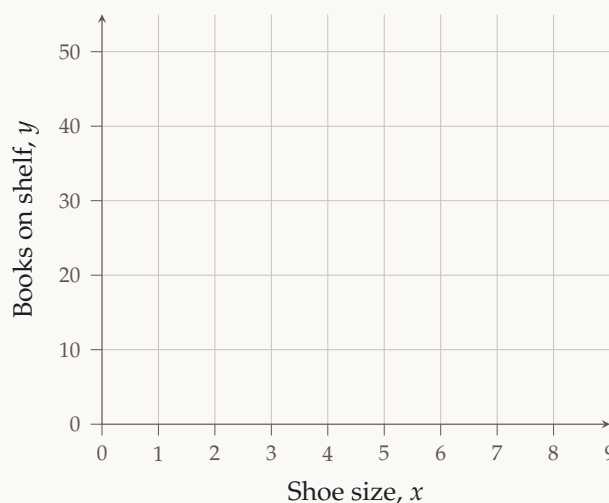
1. Your household's electricity price rises steadily at 6% a year. Estimate its doubling time.
2. World energy use has grown, on average, by about 2.3% a year. Estimate its doubling time.
3. Using your two doubling times, estimate how many times each quantity will have doubled after 60 years. Round each to the nearest whole number of doublings.

D Trend, or tangle?

Two quantities that both grow steadily over the same years will usually correlate strongly, whether or not one has anything to do with the other. The table below tracks a pupil's shoe size and the number of books on their bedroom shelf, recorded once a year.

Age (years)	10	11	12	13	14	15
Shoe size, x	3	4	5	6	7	8
Books on shelf, y	12	19	25	34	40	49

Before you calculate: Guess whether the points will scatter all over the grid, or sit close to a straight line.



1. Plot the six pairs of values from the table onto the grid above.

2. Describe the pattern your points make in one sentence. Is the correlation strong or weak, positive or negative?

3. Draw a single straight line through your points, by eye, that seems to fit them best.

4. A pupil says, "growing feet cause you to own more books." Explain what is wrong with this claim, and suggest what is really making both quantities rise together.

5. In the world energy and GDP graphs from *The Sandwich in the Middle*, plotting the raw values year by year gave an almost perfect line, but the more convincing evidence for a genuine link came from comparing how much each quantity changed from one year to the next. Using what you found with the shoe sizes and books, suggest why comparing changes is the better test.

E The accounting trick, at your school

Suppose your school needs 100 units of heat energy per day to keep its classrooms warm.

Before you calculate: Guess whether the school's reported energy consumption would rise or fall if its old gas boiler were replaced by heat pumps, assuming the classrooms feel exactly as warm either way.

1. The old gas boiler is 90% efficient, so some of the fuel's energy is lost as it burns. Calculate the gas energy needed to deliver 100 units of heat.
2. The school instead installs heat pumps with the COP you calculated in Part B, question B3 (use $\text{COP} = 3$). Calculate the electrical energy needed to deliver the same 100 units of heat.
3. Calculate the percentage fall in the school's measured energy consumption when it switches from the boiler to the heat pumps.
4. Has the actual warmth delivered to the classrooms changed at all? In a sentence, say what has really changed.

For discussion

A lecture on energy usually draws three separate things: an economist at the front, a graph on the screen, and a fuel reserve buried somewhere underground. Using your answers above, explain in two or three sentences why that picture is misleading, and where you, specifically, belong inside the diagram on the first page of this worksheet.

Answers

A. Your share of the waste

A1. $1.9 \times 10^{13} \div 8.1 \times 10^9 \approx 2300$ W per person (2 s.f.).

A2. $2300 \div 100 \approx 23.5$ times a resting body's own heat output.

A3. Accept answers noting that a person's economic footprint draws on roughly twenty times the power their own body uses, so most of the exergy used "on their behalf" is consumed by machinery, transport, heating and industry acting for them, not by their own metabolism.

B. The heat pump on your wall

B1. $2^\circ\text{C} = 275\text{ K}$; $45^\circ\text{C} = 318\text{ K}$.

B2. $\text{COP}_{\text{max}} = 318 \div (318 - 275) = 318 \div 43 \approx 7.4$.

B3. $3 \div 7.4 \approx 0.41$, so about 41% of the theoretical maximum.

B4. A heat pump does not create the heat it delivers; it moves heat that already exists in the outside air, using electrical work only to drive the pumping. Energy is conserved because most of the delivered heat came from outside, not from the electricity supply, so a COP above 1 is not a violation, merely evidence that moving heat is cheaper than making it.

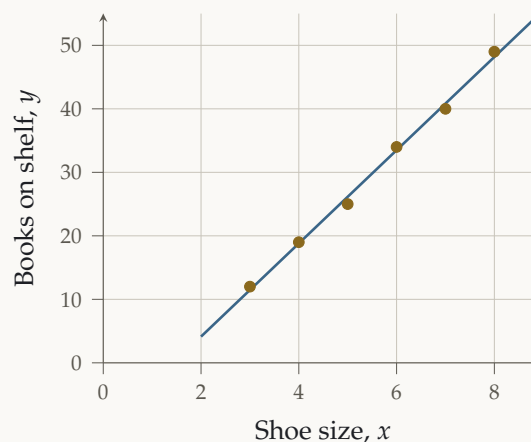
C. Compounding, at your scale and at the world's

C1. $70 \div 6 \approx 11.7$ years, roughly 12 years.

C2. $70 \div 2.3 \approx 30.4$ years, roughly 30 years.

C3. After 60 years: $60 \div 11.7 \approx 5.1$, so the bill has doubled about 5 times; $60 \div 30.4 \approx 2.0$, so world energy use has doubled about 2 times. The bill compounds far faster than world energy use, even though both are "growing steadily."

D. Trend, or tangle?



D1–D3. The six points rise from bottom left to top right in a fairly straight line, as shown above; accept any reasonable by-eye line of best fit passing close to all six points. Pupils should describe the correlation as strong and positive.

D4. Both shoe size and the number of books increase simply because the pupil is getting older each year; age is the hidden factor driving both upward, and there is no direct mechanism by which foot growth could cause book ownership. This is the same trap as reading too much into two things that both increase over time.

D5. Two quantities that are each rising steadily, for whatever reason, will nearly always trace out a strong pattern when plotted directly against each other, exactly as shoe size and books did here, so a strong pattern in the raw values proves very little. Comparing the year-to-year changes instead removes that shared upward drift, so if the changes still line up closely, as they do for world energy and world GDP, that is much better evidence that the two are genuinely connected.

E. The accounting trick, at your school

E1. $100 \div 0.9 \approx 111$ units of gas energy.

E2. $100 \div 3 \approx 33$ units of electrical energy.

E3. $(111 - 33) \div 111 \times 100 \approx 70\%$ fall.

E4. No, the classrooms receive exactly the same 100 units of heat either way. What has changed is only how much energy the meter reports, because a heat pump moves three units of heat for every unit of electricity it draws, while a boiler can deliver at most one unit of heat for every unit of fuel burned.

For discussion

Accept answers that identify the observer as embedded in the system under discussion, not standing apart from it: the economist's own body, home and institution draw on the same exergy flow as the fuel reserve and the graph they are describing, so there is no vantage point from which to watch the system without also being a small part of what it measures.